Biomedical Biceps Handbook

1. Overview

This handbook is an instructional manual that details how to build and operate the BioMedical Bicep Effort Calculator. This includes steps on how to acquire the materials, assemble and operate the hardware and software components. By using this manual, anyone should be able to build and operate the device.

2. Material acquisition

Before starting, gather these materials

MPU 5060: Buy on digikey
Arduino Nano: Buy on amazon
Athletic Wrist Band: Buy on amazon
LCD Screen: Buy on digikey
Breadboard: Buy on digikey
Wires: Buy on digikey
220 ohm Resistor: Buy on digikey
Velcro Strap: But on amazon

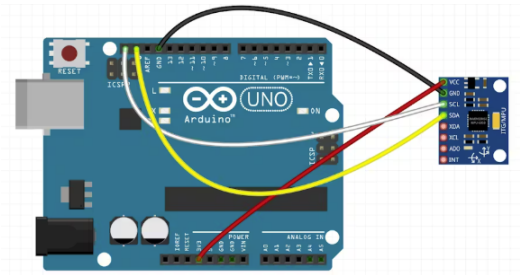
Additionally, access to a laptop and low grade sewing materials are needed to complete the project.

3. Assembly

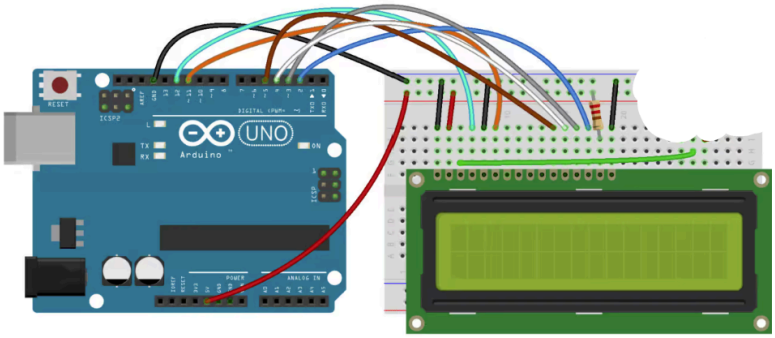
The setup of the device includes both the hardware connections of the MPU6050, LCD screen and arduino uno. Additionally, there is software setup that must be completed by the use of arduino coding software.

3.1 Hardware

Connect jumper wires to the MPU6050 pins VCC, GND, SCL, and SDA. Using the other end of the jumper wires connect to the arduino uno pins 3.3V, GNA, A4, and A5 respectively (Figure 1).



To connect the LCD screen to the arduino, a breadboard, jumper wires, and resistor must be used. First connect the LCD to the breadboard to ensure easier connections and enough ground. Connect the 5V power supply pin on the arduino to the red horizontal rail on the breadboard, and the arduino ground pin to the blue horizontal rail. Finally connect the rest of the pins with jumper writes as shown in figure 2



Finally, using an USB-A to USB-B adaptor plug in the arduino to the laptop.

3.2 Device construction

Cut each wrist band around the circumferential direction, and sow the bottom layers together. Place the LCD screen in the wristbands and sew the top layer back together. Cut a slit and push the LCD screen through the slit to expose just the screen, glue the frayed edges around the lcd screen and leave room for the wires to escape. Orient the MPU6050 so that its prongs are facing downward and are closer to your hand. Cut a small slit in the wrist band and place the MPU6050 in it, sew the slit, again leave room for the wires to escape. Tape the arduino nano to the breadboard, and use a velcro strap around the breadboard for later attachment to the subjects arm. To put on the device simply put the wrist bands on so that the LCD screen is facing upwards. Use the velcro strap to strap the breadboard/arduino just above the wrist band.

3.3 Software

The software component of the Biomedical Biceps device is responsible for reading motion data from the MPU6050 sensor, processing it with the Arduino, and displaying the calculated effort and repetition motion results on the LCD screen. The software runs through the Arduino IDE and communicates with the hardware components through specific libraries and connections.

3.3.1 Software Set up

To begin the software setup, download the arduino software (https://support.arduino.cc/hc/en-us/articles/360019833020-Download-and-install-Arduino-IDE). From the arduino library add the MPU6050 library by adafruit, the liquid crystal library, and the wire library.
Plug in the arduino to the computer usb port using a micro usb C to usb A connector.

3.3.2 Software Explanation

The software explanation describes how the Arduino code works together with the installed libraries and hardware to begin open the git link and copy the code:
https://github.com/EmileWH/BioFeedbackProject

- The MPU6050 library allows the Arduino to communicate with the motion sensor and retrieve acceleration and data that we need (effort and repetition)
- The Wire library enables I²C communication between the Arduino and the MPU6050 sensor
- The LiquidCrystal library controls the LCD display so that the processed results can be shown to the user.
- When the program runs, the Arduino continuously reads motion data from the sensor, calculates the movement related to a bicep curl, and updates the LCD display with the measured output

After uploading the code, the device operates simply by running the Arduino program and performing bicep curls while tracking the results on the LCD screen

3.3 Device operation

To operate the device, put it on as detailed in the hardware set up instructions. Next, simply press run on the arduino software. Begin curling and watch your results on the LCD screen.